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Retinal Disease Early Detection Using Deep Learning Models

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ABSTRACT Retinal diseases, which include cataracts, diabetic retinopathy, and glaucoma, cause significant numbers of cases of impaired vision throughout the world. Therefore, it is very important to be able to quickly and accurately diagnose these conditions through use of retinal fundus images to prevent permanent blindness. A new deep-learning based method is developed and presented for automated detection of multiple types of retinal disease using Convolutional Neural Networks (CNN), which will provide an improved diagnostic tool for clinicians. Four different models are utilized and compared: Custom CNN, ResNet101, EfficientNet-B3, and Swin Transformer (Swin-Tiny). An additional layer of evaluation is added by evaluating models with both intra-dataset testing and cross-dataset testing to test the models' ability to generalize from one dataset to another. The testing showed that EfficientNet-B3 achieved approximately 95% intra-dataset accuracy and a mean 4-fold cross-validation accuracy of 89.19%, difference not statistically significant with that of ResNet101 at 89.64%. In order to improve clinical interpretation of the results generated by the CNNs, the proposed system uses Explainable AI (XAI)-based visual analysis to generate heatmaps and highlight important regions. Edge quantization and model compression was also performed to understand deployment in edge devices. The proposed system demonstrated a high degree of promise for practical application in real-world retinal screening systems.

INDEX TERMS Retinal disease detection, deep learning, ResNet, EfficientNet, Grad-CAM, Explainable AI, medical image analysis.

I. INTRODUCTION

A wide variety of eye diseases including those that cause cataracts, diabetic retinopathy and glaucoma present significant global public health concerns, particularly because they frequently progress in silence until permanent damage to an individual's eyesight has been done. As a result of this silent progression, there are millions of individuals worldwide who suffer from one or all of these conditions. Individuals suffering from diabetes and elderly individuals represent two large groups at risk for developing these serious eye diseases [1] [2]. Therefore, early detection via retinal fundus imaging is critical for both the timing and effectiveness of interventions/treatment [3].

Traditionally, the identification/diagnosis of eye disease has relied on an ophthalmologist manually inspecting images. Ophthalmologic inspections of patients' eyes are extremely time-consuming, and may also be subject to substantial inter-observer variability. In addition, it is difficult for many patients living in rural areas or in areas with limited resources to obtain care from trained ophthalmic specialists [4]. These problems prompted researchers to develop computerized diagnostic systems that can automatically identify eye disease from patient's images.

Despite this progress, many prior automated screening systems have been developed and evaluated using only a single dataset, which limits confidence in how well they generalize across different imaging devices, acquisition protocols, and patient populations encountered in real-world clinical settings. Moreover, relatively few studies directly compare architectures of differing complexity under an identical experimental protocol, or pair diagnostic predictions with visual explanations that clinicians can independently verify against known disease biomarkers. Addressing these gaps motivates the three contributions of this work, described below.

The recent developments in the field of deep learning have shown a lot of promise in identifying medical classifications from medical images [5] [6]. Although many of the previous

TABLE I
CLASS-WISE DISTRIBUTION OF RETINAL FUNDUS IMAGES

Disease Class	Total Images	Training Images (80%)	Testing Images (20%)
Normal	1,074	859	215
Cataract	1,038	830	208
Diabetic Retinopathy	1,098	878	220
Glaucoma	1,007	806	201
Total	**4,217**	**3,373**	**844**

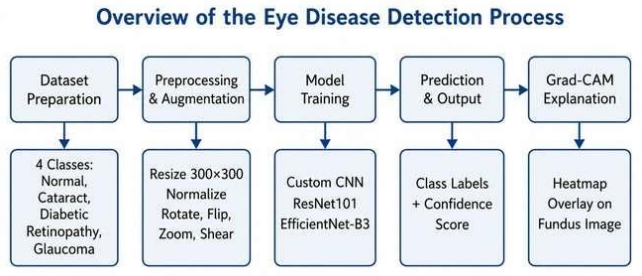


FIGURE 1. Overview of the eye disease detection process using dataset preparation, preprocessing, and model training.

approaches to identifying medical image classifications were based on deep learning models, many did not provide explanations for why certain decisions were made by a model; therefore, many lacked interpretability. Additionally, most previous studies on deep learning-based approaches to medical image classification focused solely on the results obtained using a single data set.

II. RELATED WORK

A number of recent works focus on the application of the Convolutional Neural Network (CNN) approach to various aspects of Retina Image Analysis. CNN's ability to learn its own feature set from images automatically provides an advantage over traditional methods where all relevant features had to be identified manually by a researcher prior to training a classifier [7]. In addition, the use of pre-trained models like ResNet and EfficientNet in transfer learning allows researchers to leverage the advancements achieved in image classification to improve their results on medical related tasks, especially those in which there are limitations in acquiring large amounts of labeled data [8] [9].

In medical imaging, XAI techniques are becoming increasingly important for addressing the issues of transparency and trustworthiness. One popular technique being applied to medical imaging is called Grad-CAM. It was originally designed to generate class activation maps highlighting parts of an image responsible for the model's prediction decision. This can help clinicians understand how the model arrived at a particular diagnosis or recommendation [6].

Even though there has been significant progress made toward developing CNN-based systems capable of detecting diseases in retinal images, most of the previous works evaluated system performance using a single dataset. Since

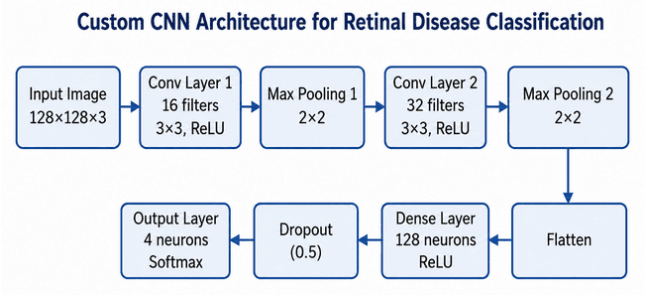


FIGURE 2. Custom CNN architecture for multi-class retinal disease classification.

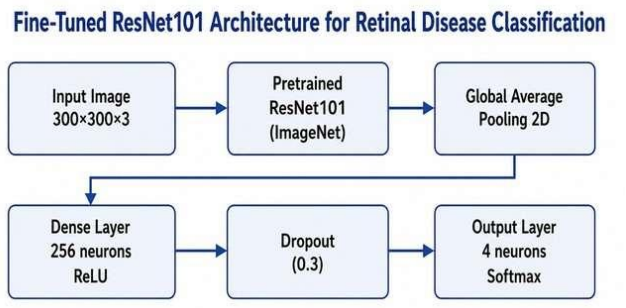


FIGURE 3. Fine-tuned ResNet101 architecture used for multi-class retinal disease classification.

the variability in terms of patient demographics and other factors associated with different datasets could affect how well a system performs in practice, a primary goal of this research effort is to overcome this limitation through three primary contributions. The first contribution involves comparing the performance of three CNN architectures differing in complexity—custom CNN, ResNet101, and EfficientNetB3—to provide a fair comparison among them. The second contribution is to apply cross-dataset validations as a means of evaluating each CNN architecture's robustness across multiple datasets acquired from different sources; this has been largely unexplored in previous literature regarding the detection of diseases in retinal images. The third contribution involves applying Grad-CAM based explanations as a method of generating visually meaningful heatmaps for disease-specific areas of the retina. Collectively, these contributions support the development of a more transparent, deployable, and generalized framework for automated retinal disease screening.

Relative to this prior work, most existing retinal/eye-disease classification studies report performance on a single dataset and a single train/test split, which leaves generalization across imaging sources largely untested; several also evaluate only one architecture, making it difficult to attribute reported gains to the model versus the dataset or preprocessing pipeline. In contrast, this study directly compares three architectures of differing complexity (custom CNN, ResNet101, EfficientNet-B3) under an identical protocol, adds 4-fold cross-dataset validation to test generalization, and pairs the classification results with Grad-

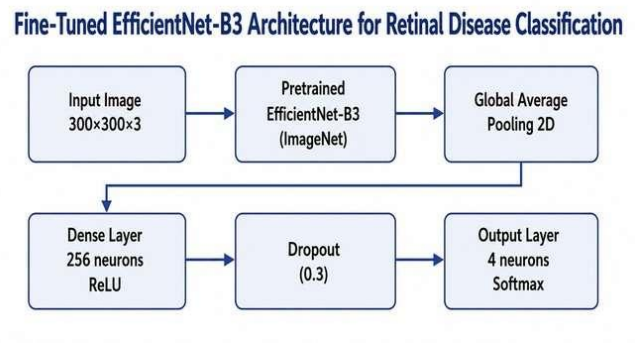


FIGURE 4. Fine-tuned EFFICIENTNET-B3 architecture used for multi-class retinal disease classification.

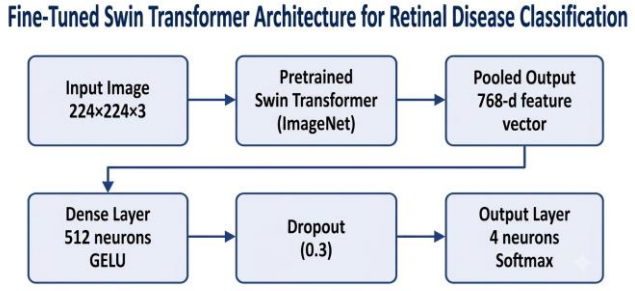


FIGURE 5. Fine-tuned Swin Transformer (Swin-Tiny) architecture used for multi-class retinal disease classification.

CAM-based visual explanations, positioning it as a more rigorous, multi-faceted evaluation than single-dataset, single-model studies in this space.

III. METHODOLOGY

A. DATASET DESCRIPTION

The system used in this paper is trained using the already existing retinal disease datasets and evaluated likewise [10]. Specifically, the intra-dataset experiments use the publicly available Kaggle "Eye_diseases_classification" dataset [21], and the cross-dataset validation experiments additionally draw on images from the Messidor [12] and IDRiD [13] public datasets to test generalization across imaging sources. The images present in the dataset are divided into four classes: Normal, Cataract, Diabetic Retinopathy, and Glaucoma [11]. The datasets used in this paper have varied resolutions, illumination, and contrast, reflecting real clinical conditions [12] [13]. The task is formulated as multi-class, single-label classification across these four mutually exclusive categories; images with coexisting pathologies are not separately represented in the source data, and extending the framework to a multi-label formulation is identified as future work. It should also be noted that the public dataset providers used in this study do not publish patient-level demographic metadata (age, sex, or ethnicity); representativeness across patient populations.

Each dataset is partitioned using a stratified 80%/10%/10% split into training, validation, and test sets. The validation set is used for early stopping and learning-rate scheduling during training, while the held-out test set (10%) is reserved solely for the final evaluation reported in Section-IV-A. This split is sufficient for training the models and ensures that there is no bias during evaluation on unseen images.

Class-wise distribution of retinal fundus images used in the study is shown in Table I.

The class distribution used in this study is moderately imbalanced, reflecting the natural relative prevalence of these conditions in the source data. The data augmentation strategy described in Section-III-B (rotation, flipping, zooming, shearing, and brightness adjustment) was applied to increase the diversity of the minority classes during training. In addition, inverse-frequency class weights were computed for each class and passed to the model during both training phases (frozen-backbone and fine-tuning) to further counteract the effect of class imbalance on the loss function; the residual effect of the imbalance on results is nonetheless discussed further in the Limitations.

B. DATA PREPROCESSING AND AUGMENTATION

To make sure that all the images across dataset are trained and tested on a consistent scale, the fundus images have been resized and preprocessed 300×300 pixels for the ResNet101 and EfficientNetB3 models. A different resolution is used for the CNN model. Pixel values are normalized to speed up convergence and have a stable training.

Implementation details: both ResNet101 and EfficientNet-B3 were implemented in TensorFlow 2.18.0 and trained on GPU hardware. Images were resized to 300×300×3 and split into training, validation, and test sets (80%/10%/10%) using stratified sampling to preserve class proportions. Both models were trained with the Adam optimizer (batch size 32) using categorical cross-entropy loss with label smoothing (0.05) and L2 kernel regularization (1×10^{-4}) on the final classification layer. Training followed a two-phase strategy: in Phase 1, the backbone was fully frozen and only the classification head was trained for 6 epochs at a learning rate of 3×10^{-4}; in Phase 2, the top 40% of backbone layers were unfrozen and fine-tuned at a reduced learning rate of 1×10^{-4} for up to 45 additional epochs (ResNet101) or 40 additional epochs (EfficientNet-B3). Early stopping (patience of 8 epochs on validation loss, restoring best weights) and a learning-rate-on-plateau schedule (halving the learning rate after 3 epochs without improvement, down to a minimum of 1×10^{-6}) were applied in both phases, and the best-performing checkpoint by validation accuracy was retained for evaluation.

To make the mentioned models more reliable, the paper used data augmentation techniques like rotating, flipping, zooming, shearing, and adjusting brightness during training as shown in Figure 1. By adding these variations, the system made the dataset more diverse and helped the models avoid overfitting. This way, the models could focus on actually learning patterns linked to diseases, instead of just picking up on quirks or artifacts specific to the training images.

C. MODEL ARCHITECTURES

The three architectures span a range of complexity and design: a lightweight custom CNN trained from scratch as a baseline, ResNet101 as an established deep residual architecture with a strong record in medical image classification, and EfficientNet-B3 as a modern, compute-efficient architecture built on compound scaling. This progression allows direct comparison of how added depth and efficiency-oriented design affect diagnostic performance on the same task. Vision Transformers and hybrid CNN-attention models were not evaluated, as their higher computational cost is a practical constraint given the dataset size and the goal of low-resource clinical deployment.

1) CUSTOM CNN

A lightweight convolutional neural network is designed as a baseline model. It consists of sequential convolutional layers with ReLU activation, max-pooling layers for spatial downsampling, and fully connected layers for classification as shown in Figure 2. The input images are rescaled to 128×128×3 pixels. The network comprises four convolutional blocks: Conv2D(16) - MaxPool, Conv2D(32)-MaxPool, producing a feature map of 8×8×64 before flattening to 4,096 units. Two fully connected layers follow: Dense (128, ReLU) with 524,416 parameters and Dense (4, Softmax) output layer with 516 parameters. The model contains a total of 557,764 trainable parameters, with zero non-trainable parameters, making it entirely trained from scratch without any pretrained weights. Although the model is computationally efficient, it has limited depth and feature-representation capacity, which is a disadvantage [14]. Its small parameter count makes it fast to train but insufficient for capturing the complex multi-scale features required for reliable retinal disease differentiation.

2) RESNET101

ResNet101 is a deep residual network pretrained on ImageNet. The model is fine-tuned by adding several important features like global average pooling and dense layers for the classification among four classes as shown in Figure 3 [15]. The input images are resized to 300×300×3 pixels. The pretrained ResNet101 backbone (42,658,176 non-trainable parameters) produces a 2,048-dimensional feature vector after Global Average Pooling 2D. Two custom head layers are appended: Dense(256, ReLU) with 524,544 parameters, Dropout(0.3), and Dense(4, Softmax) output layer with 1,028 parameters. During Phase 1, only the head is trained (525,572 trainable parameters) with the backbone fully frozen. In Phase 2, the top 40% of backbone layers are unfrozen for fine-tuning at a reduced learning rate of 1×10^{-4}. Residual connections allow effective gradient flow, which help in learning general retinal structures along with disease-specific features.

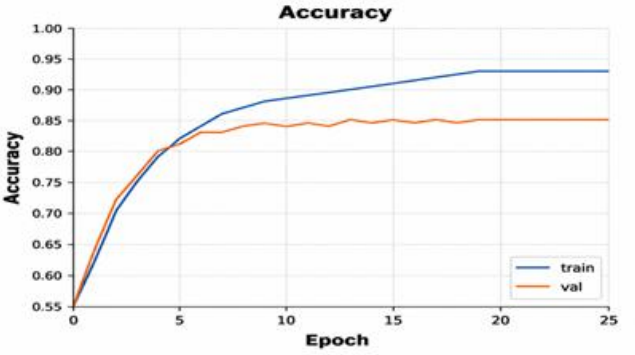


FIGURE 6. **Training and validation accuracy curves for the Custom CNN model.**

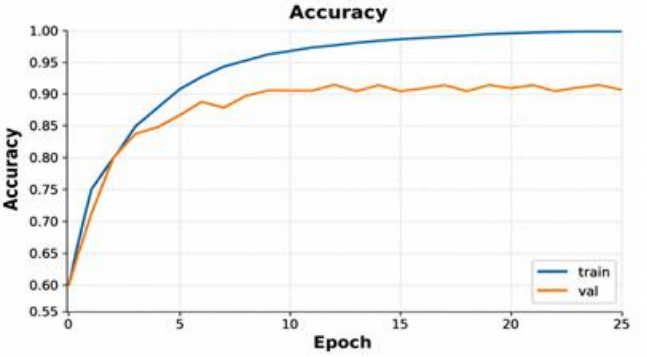


FIGURE 7. **Training and validation accuracy curves for the ResNet101.**

3) EFFICIENTNET-B3

EfficientNet-B3 is a pretrained model that balances three factors—depth, width, and input resolution—through a technique known as compound scaling shown in Figure 4. The input images are resized to 300×300×3 pixels. The pretrained EfficientNet-B3 backbone contains 10,783,535 non-trainable parameters and produces a 1,536-dimensional feature vector after Global Average Pooling 2D. The custom classification head comprises: Dense(256, ReLU) with 393,472 parameters, Dropout(0.3), and Dense(4, Softmax) output layer with 1,028 parameters. During Phase 1, only the head is trained (394,500 trainable parameters) with the backbone fully frozen. In Phase 2, the top 40% of backbone layers are unfrozen and fine-tuned at a learning rate of 1×10^{-4}. The complete model has 11,178,035 total parameters— approximately 3.9 times fewer parameters than ResNet101 — making it significantly more efficient while achieving superior accuracy. This architecture is well suited to medical image analysis, offering a balanced trade-off between performance and efficiency [16]. Fine-tuning is performed by retraining the upper layers while freezing lower-level feature extractors.

4) SWIN TRANSFORMER

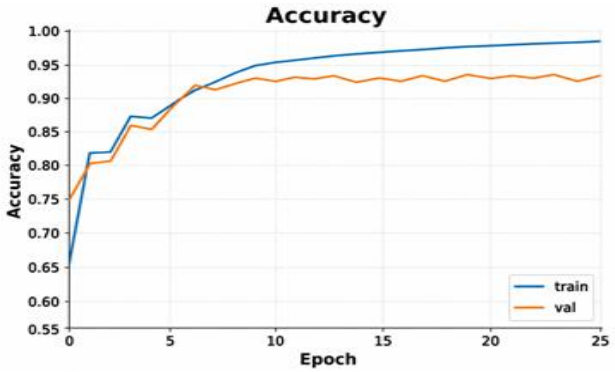


FIGURE 8. **Training and validation accuracy curves for EfficientNetB3.**

Swin-Tiny Transformer is a hierarchical vision transformer pretrained on ImageNet, evaluated here as a fourth architecture alongside the three CNN baselines as shownin Figure 5. Unlike ResNet101 and EfficientNet-B3, which accept arbitrary input resolutionsvia global average pooling, Swin's window-based attention and relative position encoding are resolution-dependent; the model was therefore evaluated at its native pretrained resolution of 224×224×3, rather than the 300×300×3 used for the CNN architectures. The pretrained Swin-Tiny backbone produces a 768-dimensional pooled feature vector, passed through a custom classification head: Dropout(0.3), Dense(512,GELU), Dropout(0.3), and a Dense(4, Softmax) output layer, as shown in Figure 5.Training followed the same two-phase strategy as the CNN architectures (frozen backbone, then partial unfreezing for fine-tuning. For Swin-Tiny, the test set was batched with a fixed batch size of 32 and the final incomplete batch (6 of 422 images) was discarded, so its metrics are computed on 416 images.

D. TRAINING STRATEGY

All models are trained using the Adam optimizer. To mitigate common issues such as overfitting and poor convergence, techniques including early stopping and learning-rate reduction are employed. Class weights are used to reduce the effect of mild class imbalance, particularly for disease categories such as diabetic retinopathy and glaucoma.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This segment of the document presents an analysis of the detection output from the model and provides a discussion on how it can be evaluated with appropriate metrics. The performance of the models will also be analyzed by use of commonly used classification metrics such as accuracy, precision, recall, F1-score, and a confusion matrix

A. INTRA-DATASET CLASSIFICATION RESULTS

The classification results for intra-dataset training where the data set was divided into 80% train, 10% validate, and 10% test (Section-III-A) are presented in this section. Accuracy percentages will present an example of what each model did well as far as its ability to generalize, learn and architecture specific.

1) CUSTOM CNN PERFORMANCE

The trained Custom CNN achieved a validation accuracy of approximately 85.9%, as shown in Figure 5. The model converged rapidly during training around epoch 10, indicating stable learning behavior for the given dataset size. However, the classification accuracy plateaued earlier than for the other architectures. This limitation is due to the model's shallow depth and limited capacity when it comes to feature representation, which reduces the model's ability to capture complex disease-specific variations present in diseases such as glaucoma and early-stage diabetic retinopathy. Despite this, the CNN model serves as a baseline for the other two architectures.

TABLE II
PAIRED PER-CLASS CLASSIFICATION METRICS FOR EFFICIENTNETB3 AND RESNET101 ON THE TEST SET (N = 422)

Class	Precision	Recall	F1-Score	Test Set	Accuracy	Model
Cataract	0.9714	0.9808	0.9761	104	—	EffNetB3
Cataract	0.9802	0.9519	0.9659	104	—	ResNet101
Diabetic Retinopathy	1.0000	0.9909	0.9954	110	—	EffNetB3
Diabetic Retinopathy	1.0000	0.9091	0.9524	110	—	ResNet101
Glaucoma	0.9000	0.8911	0.8955	101	—	EffNetB3
Glaucoma	0.8559	0.9406	0.8962	101	—	ResNet101
Normal	0.9074	0.9159	0.9116	107	—	EffNetB3
Normal	0.8727	0.8972	0.8848	107	—	ResNet101
Macro Avg	0.9447	0.9447	0.9447	422	0.9455	EffNetB3
Macro Avg	0.9272	0.9247	0.9248	422	0.9242	ResNet101

TABLE III
CROSS-VALIDATION FOLD-WISE ACCURACIES FOR EFFICIENTNET-B3 AND RESNET101

Model	Fold	Training Dataset(s)	Testing Dataset	Accuracy (%)
EfficientNET-B3	Fold-1	Folds 2, 3, 4	Fold 1	89.48
	Fold-2	Folds 1, 3, 4	Fold 2	87.48
	Fold-3	Folds 1, 2, 4	Fold 3	89.28
	Fold-4	Folds 1, 2, 3	Fold 4	90.51
	Average	–	–	89.19
	Std. Dev.	–	–	1.09
ResNet101	Fold-1	Folds 2, 3, 4	Fold 1	89.00
	Fold-2	Folds 1, 3, 4	Fold 2	88.99
	Fold-3	Folds 1, 2, 4	Fold 3	89.47
	Fold-4	Folds 1, 2, 3	Fold 4	91.08
	Average	–	–	89.64
	Std. Dev.	–	–	0.86

2) RESNET101 PERFORMANCE

The ResNet101 model achieved significantly higher intra-dataset validation accuracy of approximately 93.6%, reflecting the effectiveness of deep residual learning for retinal image classification as shown in Figure 6. The model architecture incorporates skip connections, which help maintain gradient flow and, in turn, help identify low-level and high-level disease-specific features.

ResNet101 demonstrated strong differentiation between fundus images that looked similar, particularly between normal and pathological retinal images. The deeper architecture helps the model to identify subtle structural changes in the optic disc and vascular regions [17] [18], which resulted in considerably higher accuracy than the Custom CNN. Minor misclassifications observed are primarily attributable to overlapping features.

The ResNet101 model has shown that it classifies images without overfitting and has greater generalization across image classes. Its deep residual connections helped preserve gradient flow, which helped in efficient training even with similar confusing images.

3) EFFICIENTNET-B3 PERFORMANCE

EfficientNet-B3 achieved the highest intra-dataset validation accuracy of approximately 95.0%, outperforming both the Custom CNN and ResNet101 models as shown in Figure 7.

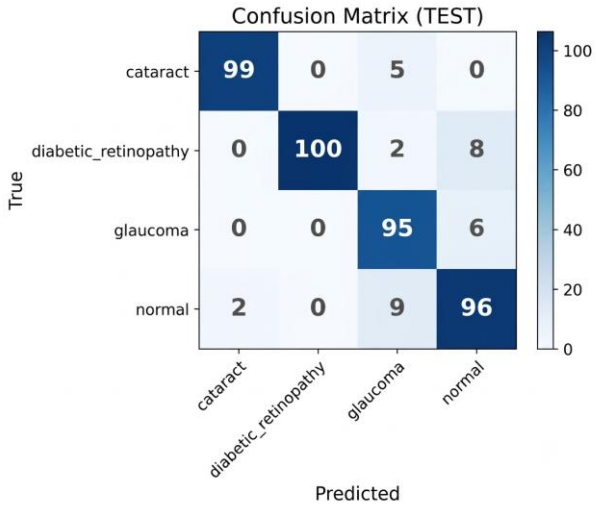


FIGURE 9. **Confusion matrix for ResNet101 on the test set.**

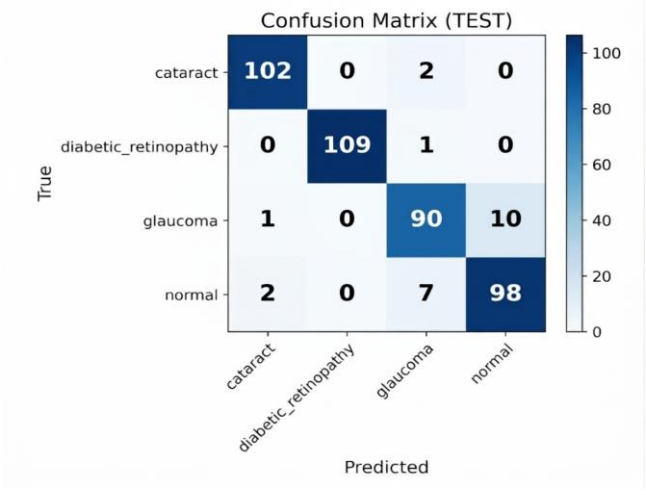


FIGURE 10. **Confusion matrix for EfficientNetB3 on the test set.**

EfficientNet's compound scaling strategy, which balances three factors—network depth, width, and input resolution—contributes to its strong performance.

This architecture helps in quickly and accurately extracting multi-scale retinal features, helping the model to differentiate fine-grained pathological patterns such as microaneurysms in diabetic retinopathy and optic cup enlargement in glaucoma [19]-[20]. EfficientNet-B3 also showed stable convergence and minimum overfitting, as seen in the closely tracked training and validation accuracy curves.

Following the highest accuracy shown by EfficientNetB3, it is the most suitable model for reliable retinal disease detection among the evaluated architectures.

The EfficientNet-B3 model is trained and tested similarly using the 80%/10%/10% split described in Section-III-A and produced the most optimal results among all three models. It is trained for 25 epochs with early stopping and learning rate

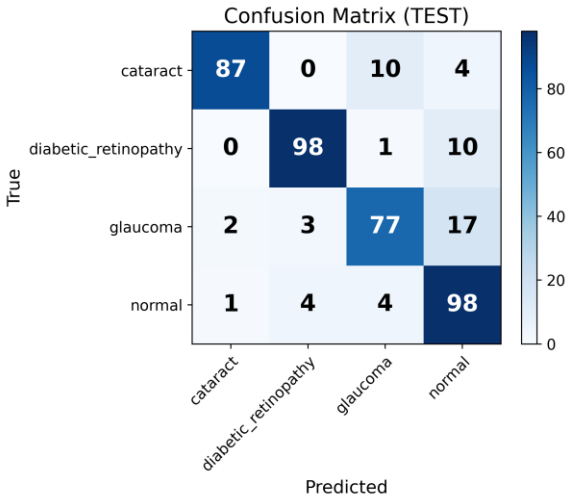


FIGURE 11. **Confusion matrix for Swin Transformer on the test set.**

reduction. The training logs show consistent incremental progress in both training and validation accuracy with a close correlation.

B. CONFUSION MATRIX ANALYSIS

The confusion matrix displays the correlation between true and predicted cases across all four diseases. A strong diagonal represents correct classification of diseases. Figure 8 shows the confusion matrix for ResNet101 on the test dataset. The higher diagonal values confirm high correct classification rates, with the most visible misclassifications occurring between the Glaucoma and Normal classes due to their overlapping and congruent structural features. Figure 9 shows the confusion matrix for EfficientNetB3 on the test set. The matrix shows considerably improved diagonal dominance compared to ResNet101, particularly for Diabetic Retinopathy and Cataract. Glaucoma and normal classes show the greatest number of false predictions, which can be due to the subtle optic disc changes in early-stage glaucoma that closely resemble normal retinal structure. The Cataracts and Diabetic Retinopathy classes show much better performance due to having more defined characteristics. These include lens opacity and lesions on the retina. Glaucoma's diagonal value was less in EfficientNet-B3 than ResNet-101. This result is logical since ResNet-101 has achieved higher Recall (.9406) for Glaucoma than EfficientNet-B3 but at the cost of lower precision (.8559). lower precision means that ResNet-101 is producing many false positives as seen in Table II. However, EfficientNet-B3 achieves the same or slightly better precision (.9000) than ResNet-101 with a slightly lower recall (.8911). Overall, both models have similar F1-scores (ResNet-101: .8962; EfficientNet-B3:.8955) so while there may be a trade-off in precision and recall, the decrease in precision is not a decline in performance.

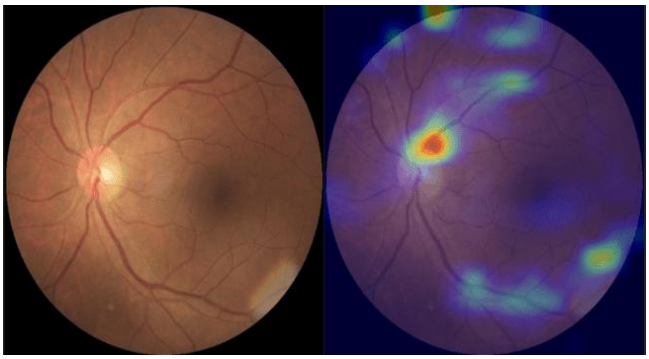


FIGURE 12. **(a) Fundus image [21] and (b) Grad-CAM confidence map overlaid on fundus image**

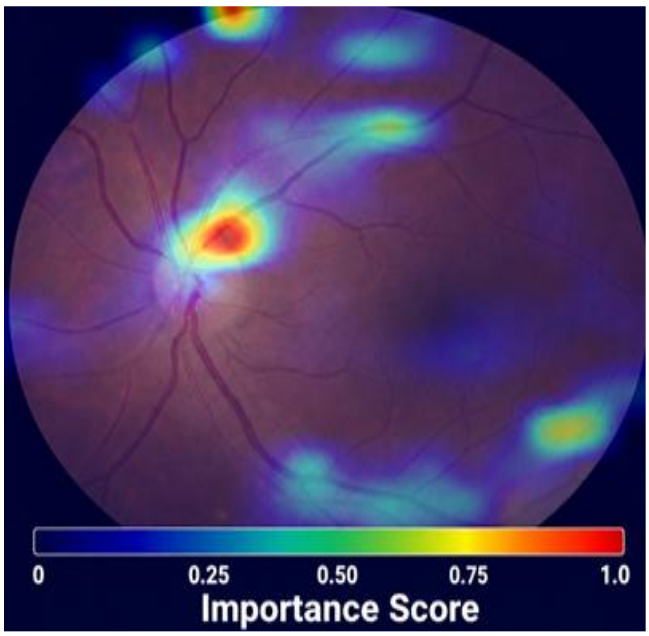


FIGURE 13. **Grad-CAM heatmap with normalized importance scale**

C. CLASSIFICATION REPORT COMPARISON: EFFICIENTNETB3 VS RESNET101

The class wise classification reports for EfficientNetB3 and ResNet101 on the test set (n=422) are shown in Table II. EfficientNetB3 achieves an overall test accuracy of 94.55%, outperforming ResNet101 which records 92.42%. F1 and precision scores are also higher for the former model across all four classes. Diabetic Retinopathy yields perfect precision 1.0000 in both models, though EfficientNetB3 achieves a superior F1-score of 0.9954 versus 0.9524 for ResNet101, caused by higher recall. For the Glaucoma class—the most challenging due to its visual similarity with normal eyes—ResNet101 attains a slightly higher recall (0.9406 vs. 0.8911), but at the cost of lower precision (0.8559 vs. 0.9000), resulting in almost similar F1-scores (0.8962 vs. 0.8955). The Normal class shows a more notable difference, with EfficientNetB3 scoring an F1-score of 0.9116 against ResNet101's 0.8848. Overall, EfficientNetB3

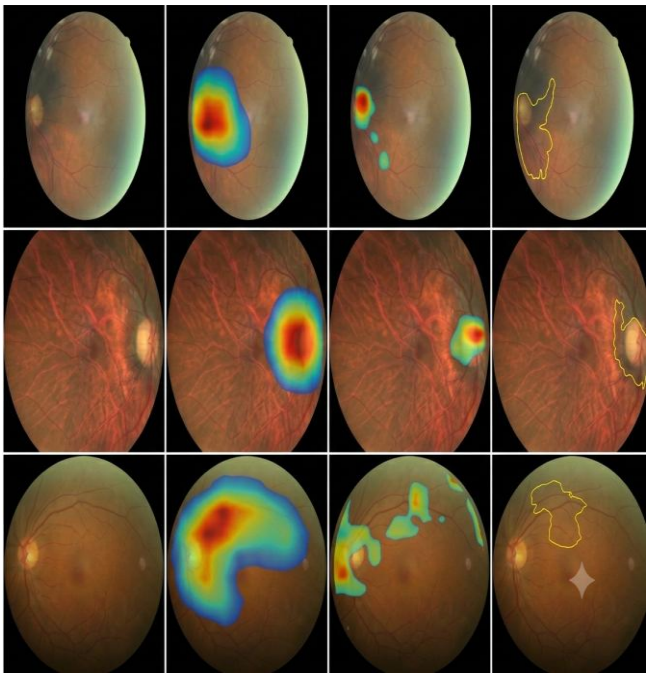


FIGURE 14. Original, Grad-CAM heatmap, SHAP and LIME explanations for representative

achieves higher macro-averaged precision, recall, and F1-score (0.9447 each) than ResNet101 (0.9272, 0.9247, and 0.9248 respectively), confirming its superior, more balanced classification performance across all disease categories. Thus, EfficientNetB3 achieves higher overall accuracy and superior macro-averaged scores across all metrics.

In addition to the metrics in Table II, Cohen's kappa and AUC-ROC (one-vs-rest, macro- and weighted-averaged) were computed for both models on the same held-out test set. EfficientNet-B3 achieved a kappa of 0.9305 and a macro/weighted AUC-ROC of 0.9909/0.9910 (per-class AUC: Cataract 0.9990, Diabetic Retinopathy 0.9999, Glaucoma 0.9798, Normal 0.9847). ResNet101 achieved a kappa of 0.8862 and a macro/weighted AUC-ROC of 0.9896/0.9897 (per-class AUC: Cataract 0.9977, Diabetic Retinopathy 0.9995, Glaucoma 0.9791, Normal 0.9820). Both models show strong agreement beyond chance and high class separability, consistent with the precision/recall/F1 already reported in Table II.

D. CROSS-DATASET VALIDATION RESULTS

To assess the generalization capability and training stability of the models, a 4-fold cross-validation experiment was conducted on the full dataset of 4,217 retinal fundus images, spanning four disease categories: diabetic retinopathy, glaucoma, cataract, and normal. In each fold, 75% of the data was used for training and the remaining 25% served as the held-out validation set, ensuring that every sample was used for evaluation exactly once. EfficientNet-B3 recorded fold-wise validation accuracies of 89.48%, 87.48%, 89.28%, and 90.51%, yielding a mean accuracy of 89.19% with a standard deviation of just 1.09% across all four folds, as summarised in Table III. ResNet101 performance was quite comparable, as it achieved fold wise accuracies of 89.00%, 88.99%, 89.47%, and 91.08%, with a mean accuracy of 89.64% and an even lower standard deviation of 0.86%, reflecting highly consistent predictions across all folds.

The low standard deviation is quite important as it indicates that neither of the architectures is overfitting to any specific part of the entire data. Also performance remains high without knowing which images are left out for testing. Both models consistently exceeded 87% accuracy in every individual fold, with EfficientNet-B3 peaking at 90.51% and ResNet101 reaching 91.08% in their respective best folds. These results consistently display that the models have learned important distinct features from the fundus images rather than just memorizing training samples. The close similarity between fold results and the overall intra-dataset test accuracy shows that the experimental setup is accurate and efficient and the quality of learning is good.

This relatively stable cross-validation performance is attributed in part to the preprocessing pipeline (consistent image resizing and pixel normalization, Section-III B) and the augmentation strategy, both of which reduce sensitivity

TABLE IV
PAIRED T-TEST COMPARING EFFICIENTNET-B3 AND RESNET101 ACROSS THE FOUR CROSS-VALIDATION FOLDS (N=4, DF=3).

Metric	EfficientNet-B3 (mean)	ResNet101 (mean)	Paired t-test (p-value)
Accuracy	88.81%	89.42%	0.50
Cohen's kappa	0.8506	0.8548	0.73
AUC-ROC (macro)	0.9832	0.9850	0.46

TABLE V
EFFICIENTNET-B3 TEST-SET PERFORMANCE UNDER SYNTHETIC IMAGE-QUALITY DEGRADATION (N=422).

Degradation	Accuracy	Macro F1	Kappa	Accuracy Drop
Clean (baseline)	94.31%	0.9427	0.9241	—
Blur (mild, r=2)	81.75%	0.8071	0.7566	12.56 pts
Blur (severe, r=5)	57.58%	0.5373	0.4339	36.73 pts
Noise (mild, std=15)	74.41%	0.7420	0.6586	19.91 pts
Noise (severe, std=35)	41.47%	0.3443	0.2213	52.84 pts
Low illumination (0.6×)	91.94%	0.9189	0.8925	2.37 pts
Low illumination (0.3×)	83.89%	0.8324	0.7849	10.43 pts
JPEG artifact (q=20)	66.11%	0.6249	0.5508	28.20 pts

TABLE VI
EDGE QUANTIZATION BENCHMARK OF THE EFFICIENTNET-B3 TENSORFLOW LITE MODELS

Metric	FP32 (mean)	FP16 (mean)	Pruned FP32 (mean)	Pruned INT8 (mean)
Model size (KB)	43276.6	21730.4	43276.6	12619.7
Gzip size (KB)	—	—	—	7676.6
Size reduction (raw)	—	49.8%	—	70.8%
Test accuracy (%)	93.13	93.13	90.05	86.97
Macro F1	0.9303	0.9303	0.9980	0.8654
Avg. latency (ms)	71.47	73.12	42.40	37.83
P95 latency (ms)	74.75	79.36	44.90	42.44
Accuracy drop (pts)	—	0.00	—	3.08

to resolution and illumination differences across the combined image sources. Differences in acquisition protocol or unmeasured demographic differences across the source datasets cannot be fully ruled out as contributors to the residual fold-to-fold variance observed.

In addition to accuracy, Cohen's kappa and AUC-ROC (one-vs-rest, macro-averaged) were computed for each cross-validation fold reported in Table III. For EfficientNet-B3, per-fold kappa was 0.8469, 0.8215, 0.8645, and 0.8696 (mean 0.8506, std 0.0188), and per-fold macro AUC-ROC was 0.9788, 0.9797, 0.9870, and 0.9875 (mean 0.9832, std 0.0040). For ResNet101, per-fold kappa was 0.8419, 0.8544, 0.8455, and 0.8772 (mean 0.8548, std 0.0137), and per-fold macro AUC-ROC was 0.9820, 0.9864, 0.9838, and 0.9879 (mean 0.9850, std 0.0023). Both models therefore show consistently strong agreement beyond chance and high class separability across all folds, in addition to the accuracy already reported in Table III.

In Table IV paired t-test comparing the two models across the four folds, found no statistically significant difference in accuracy ($t=-0.76$, $p=0.50$), kappa ($t=-0.37$, $p=0.73$), or macro AUC-ROC ($t=-0.85$, $p=0.46$), indicating that the small numeric gap between EfficientNet-B3 and ResNet101 in cross-validation is not reliably distinguishable from fold-to-fold variance at this sample size ($n=4$ folds); this small-n limitation is itself noted, as a paired test over only four folds has limited statistical power. No metric shows a statistically significant difference at $p<0.05$.

E. GRAD-CAM-BASED VISUAL ANALYSIS

Grad-CAM visualizations [6] display the decision-making process of the model and help medical professionals to confirm their analysis by highlighting critical features. The heatmaps consistently highlight clinically relevant retinal regions, including lesion clusters in diabetic retinopathy, lens opacity in cataract cases, and optic disc regions in glaucoma images. Normal retinal images show diffuse and low-intensity activations, indicating the absence of pathological features.

These observations confirm that the models rely on medically relevant features in the retina rather than background features to help identify the diseases.

To validate the system's explainability, Grad-CAM is used to visualize the regions of the retina most influencing the model's decisions.

The Grad-CAM photos highlight specific features such as lesions, hemorrhages, or optic nerve changes, depending on the class.

- Cataract: High activation near the lens opacity region.
- Diabetic Retinopathy: Targeted highlights around hemorrhagic and microaneurysm zones.
- Glaucoma: Attention concentrated on the optic disc and cup region.
- Normal: Diffuse, low activation across the retinal surface.

Figure 10 illustrates a representative Grad-CAM confidence map overlaid on a diabetic retinopathy fundus image. The figure highlights specific regions or zones which helped the model to classify the disease. The colors red and yellow indicate higher activation compared to blue and green, which indicate lower activation. Grad-CAM is used as a tool that helps identify disease-relevant regions, rather than functioning as a diagnostic model itself. The heatmap overlaid on the eye image helps verify whether the model is focused on medically relevant features rather than noise. The brighter the portion of the heatmap the more significant it is in classifying and identifying the disease. Red/yellow represents important and blue or green represents not so important zones.

Figure 11 shows the Grad-CAM color scale ranges from blue (0.0 – negligible activation) through green and yellow to red (1.0 – peak importance). The areas of the image where the confidence exceeds 0.7 are defined as High Confidence Diagnostic Regions and they are typically located at well-defined clinical markers; microaneurysms and hemorrhages (Diabetic Retinopathy), lens opacities (Cataract), cup/disk margins (Glaucoma) and diffused low activation patterns (Normal). These visualizations allow physicians to see if their models have paid attention to clinically relevant features for diagnosing eye disease in an unambiguous way, and trustworthily. It is important to note that these interpretations were made by comparing the heat maps to the clinically identified biomarkers described in cited literature [6] and not through a blinded validation study using ground truth annotations created by practicing ophthalmologist which is medically unfeasible due to approvals.

F. ROBUSTNESS TO IMAGE QUALITY DEGRADATION

Retinal screening images from real-world environments typically exhibit degradation due to poor illumination, blur, noise, and compression artifacts. Therefore, synthetic degradations were applied to evaluate model performance under conditions of varying severity including Gaussian

blur, Gaussian sensor noise, reduced illumination and JPEG compression on the held-out test set (n = 422). Additionally, an extension of this evaluation to ResNet101 is identified as a potential future step. The results of this evaluation are shown in Table-V. The model demonstrated robustness to lighting variation with only a drop of 2.37 points under moderate darkening, and 10.43 points under severe darkening. Thus, it appears that the brightness-augmentation strategy used during training transferred well to this condition. However, the model was less robust to blur and sensor noise which resulted in larger drops in accuracy at higher levels of severity (up to 36.73 and 52.84 respectively) alongside corresponding kappa reductions under severe noise.

G. EDGE QUANTIZATION AND MODEL COMPRESSION RESULTS

To assess suitability for future deployment on resource-constrained edge devices, the trained EfficientNet-B3 model was converted to TensorFlow Lite using post-training edge quantization and compression, with no retraining. Four variants were produced: an FP32 TensorFlow Lite baseline; an FP16 model, in which the weights are quantized to 16-bit floating point (default converter optimizations with float16 as the supported type); an INT8 model obtained by full-integer post-training quantization, calibrated on a representative set of 200 randomly sampled training images (no test images were used), restricted to INT8 built-in operators with int8 input and output tensors; and a pruned variant, in which weight pruning was applied before conversion to FP32 and INT8. All variants were evaluated on the same stratified held-out test set as the other models (n = 422), at 300×300 input resolution with EfficientNet preprocessing. Inference was run one image at a time (batch size 1) on CPU through the TensorFlowLite interpreter with the XNNPACK delegate, and latency is reported as the mean and 95th percentile (P95) of the per-image interpreter invocation time, excluding image loading and preprocessing. Because these measurements were taken on a generalpurpose CPU rather than on an embedded device, they indicate relative rather than absolute edge performance.

Table VI summarises the two compression pipelines. Converting the model to FP16 halved its size (43,276.6 KB to 21,730.4 KB, a 49.8% reduction) with no change in test accuracy (93.13%) or macro F1-score (0.9303), although average CPU latency rose slightly (71.47 ms to 73.12 ms; P95 74.75 ms to 79.36 ms). This is likely because the CPU has no native FP16 arithmetic, so the weights are converted back to FP32 at inference and FP16 benefits storage rather than speed on this hardware. by particular, the pruned FP32 model was itself 3.08 points below the unpruned FP32 model (93.13%), so the cumulative loss from pruning and quantization of INT8 is approximately 6.2 points and 3.08 with the pruned FP32 model(90.05%). FP16 conversion is therefore the lossless option for edge deployment, whereas pruning with INT8 quantization offers the smallest footprint and fastest inference where a moderate accuracy trade-off is acceptable. Latencies from the two pipelines were obtained in separate benchmark runs and should be compared within, not across, pipelines, and the FP32 TFLite accuracy (93.13%) is slightly lower than the Keras test accuracy in Table II (94.55%), so comparisons should be made within this benchmark. Validation on physical edge hardware and a class-wise analysis of the effect of quantization, particularly on Glaucoma, are left for future work.

V. DISCUSSION AND LIMITATIONS

Considered together, the results support EfficientNet-B3 as the strongest-performing architecture evaluated, with ResNet101 close behind and no statistically significant difference between the two on cross-validation metrics (Section-IV-D). This intra-dataset accuracy (95.0%) compares favorably with recent EfficientNet-based classifiers addressing the same four disease categories, such as the 93.8% accuracy reported in [22], which did not include cross-dataset validation or explanation-based interpretability. It also compares favorably with the 94.88% best accuracy reported in [23] for EfficientNet across three retinal disease categories using a single dataset and 10-fold cross-validation, without generalization testing across independently sourced datasets. The cross-dataset and Grad-CAM components of the present protocol therefore extend beyond what is typically reported in this line of work, representing a more rigorous test of generalization than single-dataset evaluation alone.

Beyond these findings, a number of scope limits also merit discussion, many of them related to practical constraints of working with publicly available benchmarks rather than limitations of the research methodology itself. The three datasets used here - Kaggle Eye_diseases_classification, Messidor, and IDRiD - do not provide patient-level demographic information, a common practice in the publication of de-identified public data; as a result, representation across age, gender, race/ethnicity, and imaging equipment cannot be directly verified. The datasets are also not newly collected (Messidor, 2014; IDRiD, 2018); although both remain widely used, well-validated benchmarks in this research area, revalidation against newer imaging equipment is identified as a natural next step. In addition, there is some moderate imbalance in the classes within each dataset, which is expected and is mitigated using data augmentation and inverse-frequency class weighting.

Due to the fact that prospective clinical evaluation and comparison with practicing ophthalmologists would require an institutional IRB approval, a comprehensive review of a clinical dataset, and formal approval from practicing ophthalmologists prior to conducting any comparisons, it

was beyond the scope of this dataset-based study. Clinical application is a logical follow up. The performance of EfficientNet-B3 was also evaluated over degraded image quality. The results indicated that the model had good resilience to variations in illumination, but greater sensitivity to degradation caused by blur and sensor noise. Clinical biomarkers were examined to assess Grad-CAM visualizations patterns consistent with those observed at known biomarker locations. Consistent and clinically meaningful localization was noted. A blinded clinician rating study was not medically feasible and is future work. Although inference latency and memory footprint on constrained hardware were not within the present scope, they should represent a relatively simple extension once a deployment platform has been selected.

VI. CONCLUSION

The research evaluated the performance of three deep learning networks (models) in the detection of retinopathy using fundus image analysis. The top performing model was EfficientNet-B3 which reached an accuracy of 95% when testing within the same dataset and 89.19% when tested with different data splits as part of four-fold cross-validation. In terms of overall performance, ResNet-101 ranked second, followed by a simpler custom convolutional neural network (CNN). Although the custom CNN trained faster than both of the other two models, it did not have sufficient power to differentiate between visually similar classes of disease.

Grad-CAM visualizations revealed that the models were focusing on the right parts of the eye — lesion clusters in diabetic retinopathy, the optic disc in glaucoma, and lens regions in cataract cases — rather than picking up on irrelevant background details. This makes the system not only accurate but also transparent enough for clinicians to trust. With future work on making these models lighter for low-resource settings, expanding the range of diseases covered, and training across multiple hospitals without sharing patient data, this kind of tool could realistically support retinal disease screening in everyday clinical practice.

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